

# **A Bayesian Approach to Handling Noisy Labels: The Case of Occupational Coding**

## **Objective**

The aim of this thesis is to investigate how to deal with uncertain or conflicting classifications (*noisy labels*) in datasets where the “true” category is unobserved. Such situations frequently arise in domains such as medical diagnostics, text annotation, or automated occupational coding.

The thesis also outlines application areas for these methods: originally developed in medical diagnostics, they are increasingly used in machine learning, text annotation, and labeling tasks for large language models (LLMs). This broader context will be briefly introduced in the first section. The work will reflect on the opportunities that probabilistic models offer for these emerging applications, such as transparency, uncertainty quantification, and targeted relabeling, as well as on existing methodological or practical challenges.

The central question is how probabilistic models can be used to statistically estimate both the underlying true class and the quality of individual raters. The model is based on the Dawid & Skene (1979) approach, which simultaneously models the latent “true” category and the individual error structures of raters. The thesis will examine both the classical maximum likelihood variant (via the EM algorithm) and a Bayesian extension that explicitly incorporates uncertainty and prior information. In addition, alternative approaches to handling such data structures will be reviewed and compared (literature review).

Using the example of semi-automated occupational coding, the thesis will demonstrate how this model can be applied and interpreted in practice. The focus is not on substantive domain analysis but on a methodological illustration of how probabilistic models can be used to quantify uncertainty in the absence of ground truth and where potential opportunities and limitations arise in their application.

## **Methodological Section**

The methodological section forms the core of the thesis. It begins with an overview of existing methods for modeling noisy labels, focusing on the classical Dawid & Skene approach as a maximum likelihood estimation method (via the EM algorithm). This model describes the joint probability of observed rater labels given an unobserved latent “true” class and allows estimation of both rater errors (confusion matrices) and class prevalences.

Building on this, the Bayesian modeling framework will be introduced, extending the same structure to incorporate uncertainty and prior knowledge about rater quality and class distributions. This allows full quantification of uncertainty in the model, both regarding the true class and the model parameters, via posterior distributions. Mathematically, the relationship between likelihood, prior, and posterior will be explained, along with an accessible introduction to sampling algorithms (in particular, MCMC and NUTS).

The assumptions and limitations of these models will also be discussed.

An extension question will be considered: Can additional covariates be incorporated (e.g., respondents' education level)? Does the ground truth depend on them?

This raises two distinct research directions:

(1) how raters differ, and

(2) how the predicted ground truth can be determined and interpreted.

Possible extensions such as handling missing data may also be discussed.

## **Data Basis**

The empirical example uses a dataset of approximately 1,400 items (including missing values), each classified multiple times: two manual codings and one automated assignment (for both KldB and ISCO occupational groups). This dataset provides a realistic scenario with multiple raters and no known true labels, making it well suited for demonstrating probabilistic label models.

## **Methodological Procedure in the Application Section**

The application section begins with a descriptive analysis of the ratings. Next, the Bayesian model is implemented in Stan.

The Bayesian variant enables estimation of posterior distributions for the prevalences of true classes, the rater quality parameters, and the posterior probabilities of the true class for each item.

The models will then be compared in terms of stability, interpretability, and usefulness for uncertainty quantification.

Initial analyses showed that the model converged only for a maximum of 2 categories instead of 10. Building on this, it is investigated what is needed for the sampling algorithms to converge. Further data can be simulated for this purpose.

## **Expected Contribution**

The thesis provides a methodological contribution to modeling classification data with uncertain or missing ground truth. It demonstrates how uncertainty regarding labels, rater quality, and class distributions can be represented within a probabilistic framework and illustrates this with a real-world occupational coding dataset.

The result is a transferable modeling approach applicable across a wide range of fields, from social science coding to medical diagnostics and machine learning/NLP, whenever labels are noisy, inconsistent, or ambiguous.